How Much Accuracy Is Worth the Additional Computing Cost? Investigating the Trade-off Between AI Performance and Computational Resources

A. Introduction

As artificial intelligence systems continue to grow in size, a critical question remains inadequately addressed in current practice: how much additional computational effort is justified by a given improvement in accuracy? It is well established that increasing model size, dataset size, and computation can improve AI performance. However, it is less clear whether a given improvement in accuracy is worth the additional resources required to achieve it.

This is a significant problem because increasing computational effort requires additional resources, including computing time, energy consumption, and infrastructure. Yet AI development has often treated accuracy as the primary measure of progress, with less attention given to the resources required to achieve a particular improvement. Therefore, this study asks: How does the accuracy of an AI model change when the amount of computational effort is increased, and does the improvement in accuracy become smaller at higher levels of computation?

B. Scope of the Problem

This problem is important because AI systems are becoming more demanding in terms of computational resources. Modern AI models can require significant computing power. Schwartz et al. show that the computational requirements of deep learning systems have increased substantially. They also explain that if we focus only on accuracy, we may overlook the resources and environmental impact required to improve model performance.

Computational resources can be required at different stages of AI development. They can be used to train the model, process large datasets, run multiple experiments, and test different settings. They can also be required when using the model for inference. Schwartz et al. explain that factors such as dataset size, the cost of processing examples, and the number of experiments can contribute to the overall computational requirements.

Because of this, similar levels of performance may be obtained using different amounts of computational resources. As computational effort increases, the improvement in accuracy may

also become progressively smaller. Therefore, there is a need to examine the relationship between model accuracy and the computational effort required to achieve it, and to determine whether additional computational effort continues to provide meaningful improvements in performance.

C. Literature Review

Scaling Laws and Computational Resources

Kaplan et al. (2020) showed that AI performance is related to model size, dataset size, and training compute, establishing that computational resources can be treated as a measurable factor affecting model performance. Hoffmann et al. (2022) further showed that computational resources can be allocated more efficiently by balancing model size and training data. Together, these studies establish that the amount and allocation of computation have a strong effect on AI performance. However, their main focus is on determining efficient scaling strategies rather than directly measuring the marginal benefit obtained from additional computation after a model configuration has been established.

Accuracy Versus Efficiency

Schwartz et al. (2020) argued that accuracy should not be the only measure used to evaluate AI systems and proposed the Green AI framework, which emphasizes computational efficiency alongside model performance. The study highlights the problem of diminishing returns, where increasingly larger computational resources may produce progressively smaller improvements. However, the framework mainly provides recommendations for reporting and evaluating efficiency rather than a specific experimental method for determining when an additional improvement in accuracy is worth its computational cost.

Energy and Environmental Cost

Strubell et al. (2019) demonstrated that computational requirements for AI can also have financial and environmental consequences. Their work shows that training AI systems can require significant energy and resources, making computational efficiency an important practical consideration. However, their cost estimates depend on the hardware and models available at the time of the study and do not directly establish how the value of an accuracy improvement changes with additional computation.

Emerging Work on Marginal Returns to Compute

Recent research has begun to examine diminishing returns more directly. Shukla (2025) modeled the relationship between model accuracy and training compute and found that accuracy gains become smaller as computational resources increase. Jin et al. (2025) extended this type of analysis to inference-time computation and also identified trade-offs between

additional computation, accuracy, and energy use. These studies provide useful evidence that marginal returns can be quantified, but they focus on particular benchmarks or specific stages of AI computation.

Research Gap

Previous studies have established that additional computation can improve AI performance and that computational costs can increase substantially. However, existing research has focused mainly on scaling strategies, efficiency recommendations, or specific training and inference settings. There is still a need for a simple quantitative framework that directly compares the additional accuracy obtained from increasing computational resources with the additional computational cost required to obtain that improvement. The present study addresses this gap by experimentally measuring marginal accuracy gain and marginal efficiency across different computational budgets.

D. Objectives

The objectives of this study are:

1. To study the relationship between AI model performance and computational cost using both experimental results and existing results from published studies and public benchmarks.
2. To compare the accuracy obtained at different levels of computational resources.
3. To calculate and compare changes in accuracy and computational cost across different models and computational levels where comparable data are available.
4. To observe whether the improvement in accuracy becomes smaller as computational resources are increased.

E. Methodology

Research Design

This study will use a quantitative approach combining a small-scale experimental study with analysis of existing results. A small number of selected AI modelsl will be trained using a suitable publicly available dataset, and its performance will be measured at different levels of computational effort. In addition, relevant results from existing studies and public benchmarks will be collected and analyzed to provide further comparison and context.

Experimental Procedure

A small number of selected AI models and a suitable dataset will be used for the experiment. The models will be trained for different numbers of epochs, representing increasing levels of computational effort. Other major conditions, such as the model architecture, dataset, and

training settings, will be kept the same as far as possible so that the effect of increasing computational effort can be observed.

For each training level, the total training time and model accuracy will be recorded. In addition, relevant accuracy and computational cost results from existing studies and public benchmarks will be collected where the reported conditions are sufficiently comparable. These results will be analyzed separately and used to compare and contextualize the experimental findings.

Data Collection

The primary experimental data will be collected from the AI models trained as part of this study. For each experimental model, relevant performance and computational measurements will be recorded. Secondary data will be collected from published studies and public benchmarks that report AI model performance and computational requirements. The collected results will be organized according to the model, task, performance measure, and computational cost measure to allow comparison where the reported values are sufficiently comparable.

Measurement

The main variable being changed in the experiment will be the amount of computational effort used for training. Computational effort will be measured by the number of training epochs. The main performance measure will be the accuracy of the AI model on the test set.

For each experiment, the following values will be recorded:

- Number of training epochs
- Total training time
- Model accuracy on the test set
- Change in accuracy from the previous training level

Total training time will be used as a practical measure of the computational resources required for each training level. The same hardware and major experimental settings will be used for all experimental conditions as far as possible so that the measured results can be compared consistently.

Procedure for Comparison

The experiment will begin with a lower number of training epochs and then use progressively higher numbers of training epochs. The accuracy and training time obtained at each level will be compared with the previous level.

This comparison will be used to determine whether increasing computational resources continue to provide similar improvements in accuracy or whether the improvement becomes smaller at higher computational levels.

F. Plan for Analysis

The experimental results and the selected results from existing studies and public benchmarks will first be organized according to model performance and the available computational measures. Results will be compared where the task, performance measure, and computational measure are sufficiently comparable.

For the experimental results, a graph of accuracy against training time will be used to visualize the relationship between model performance and the resources required for training. This will help show whether increasing computational effort continues to provide similar improvements in accuracy or whether the improvement becomes smaller at higher levels of computation.

The change in accuracy between two consecutive computational levels will be calculated using:

Marginal Gain = Accuracy□ − Accuracy□₋₁

The change in accuracy will also be compared with the additional training time using:

Marginal Efficiency = (Accuracy□ − Accuracy□₋₁) / (Training Time□ − Training Time□₋₁)

Here, training time is used as a practical measure of the additional computational resources required during the experiment. The marginal gain and marginal efficiency values will then be compared across the different experimental training levels. A decrease in marginal gain or marginal efficiency at higher computational levels will indicate that additional computational effort is producing smaller improvements in accuracy. The results from existing studies and public benchmarks will be analyzed separately using the computational measures reported by the respective sources. These results will be used to provide additional context and to compare the observed accuracy–compute relationship where the reported conditions are sufficiently comparable. Overall, the analysis will be used to understand the accuracy–compute trade-off and to identify whether diminishing returns are present in the experimental results and supported by the comparable secondary results.

G. Limitations

1. **Limited experimental scale:** The primary experiment will use only a small number of lightweight AI models due to the limited computational resources and time available for the study. Therefore, the experimental findings may not represent the behavior of larger or more complex AI systems.

2. **Dataset and task dependence:** The experimental results will depend on the selected dataset, task, model architecture, and training settings. The observed accuracy–compute relationship may therefore differ for other datasets, tasks, or models.
3. **Hardware dependence:** Training time is used as a practical measure of computational cost in the experiment. Training time can vary depending on the hardware and software environment, so the measured values may not be directly comparable with results obtained on different systems.
4. **Limited comparability of secondary data:** Results collected from published studies and public benchmarks may use different models, datasets, hardware, performance metrics, and definitions of computational cost. Therefore, secondary results will only be compared directly when the reported conditions are sufficiently comparable.
5. **Accuracy as the main performance measure:** The study primarily uses test accuracy to measure model performance. Other factors such as inference latency, memory usage, robustness, reliability, and energy consumption are not the main focus of the experimental analysis.
6. **Limited availability of computational-cost information:** Some published studies or benchmarks may report accuracy without providing sufficient information about training time or computational resources. Such results may therefore be useful for contextual analysis but may not be suitable for calculating marginal computational efficiency.

Conclusion

This study investigates the trade-off between AI model performance and the computational resources required to achieve that performance. While increasing computational effort can improve model accuracy, the improvement may become smaller as more computational resources are used. Understanding this relationship is important because increasing computational effort requires additional computing resources and time.

To investigate this trade-off, the study uses a quantitative approach combining a small-scale experimental study with analysis of existing results from published studies and public benchmarks. In the experimental component, a small number of lightweight AI models will be trained using a suitable publicly available dataset at different levels of computational effort. The number of training epochs will represent computational effort, while total training time will be used as a practical measure of the resources required for training. Model accuracy on the test set will be recorded at each training level.

In addition, existing results from published studies and public benchmarks will be collected and analyzed where the reported model performance and computational measures are sufficiently

comparable. These secondary results will provide additional context and allow the accuracy–compute relationship to be examined beyond the small-scale experimental setup.

The collected results will be analyzed by comparing changes in accuracy with changes in the measured computational resources. Marginal gain and marginal efficiency will be calculated to determine whether additional computational effort continues to provide similar improvements in accuracy or whether the benefits decrease at higher computational levels. The study will therefore focus on identifying the presence of diminishing returns in the relationship between AI performance and computational effort.

Although the study is limited by the small experimental scale, hardware constraints, and differences between secondary sources, it aims to provide a practical quantitative view of the accuracy–compute trade-off. The findings can help illustrate that increasing computational effort does not necessarily result in proportionally higher accuracy, and that the value of additional computation should be considered alongside the performance improvement it provides.

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